

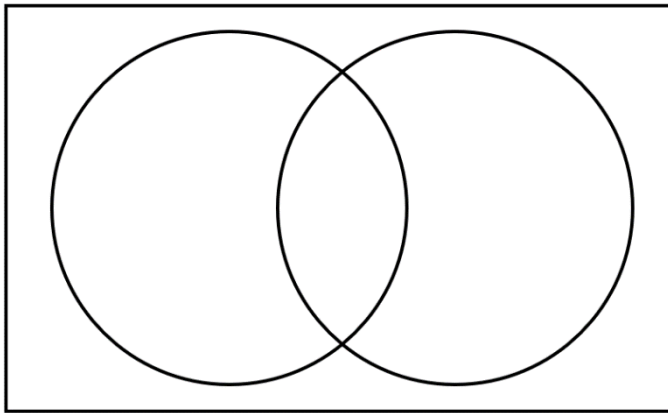
Unit 4: Probability, Random Variables, and Probability Distributions
HW 3 Homework – Venn Diagrams, Unions, and Intersections

Name: _____

1) At a liberal arts college in the Midwest, 39% of freshman are enrolled in both a math course and an English course, 11% are enrolled in a math course, but not an English course, and 63% are enrolled in an English course.

a) Construct a Venn diagram to illustrate this situation. Be sure to label all sets.

b) What is the probability that a freshman student selected at random is taking an English course but not a mathematics course?

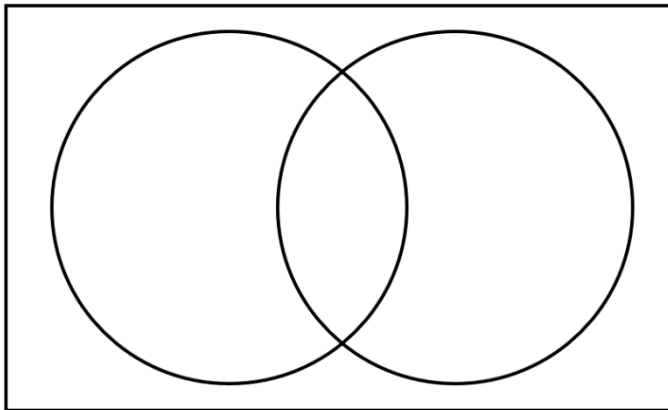


c) What is the probability a student is not enrolled in either course?

2) At that same liberal arts college, 56% live in a campus residence hall, 62% participate in a campus meal program and 42% do both.

a) Construct a Venn diagram to illustrate this situation. Be sure to label all sets.

b) What is the probability that a randomly selected student either lives or eats on campus?



c) What is the probability that a randomly selected person lives off campus and doesn't have a meal program?

3) Create a table to display all data that is describes.

M = Male

F = Female

D = owns a desktop

L = owns a laptop

B = owns both

N = owns neither

$$P(F \cap D) = \frac{21}{235}$$

$$P(M) = \frac{153}{235}$$

$$P(L) = \frac{106}{235}$$

$$P(M \cap L) = \frac{52}{235}$$

$$P(F \cap N) = \frac{6}{235}$$

$$P(N) = \frac{10}{235}$$

$$P(F \cup D) = \frac{86}{235}$$

	Desktop	Laptop	Both	Neither	Total
Female					
Male					
Total					

4) 162 students are receiving awards at an awards banquet. 48 receive math awards, 78 receive English awards, and 14 receive both Math and English awards. What is the probability that a randomly chosen student has won an award for English or Math?

5) The probability of rain on Saturday or Sunday is given as 0.5. The probability of rain on Saturday is 0.2 and the probability of rain on Sunday is 0.4. What is the probability that it will rain on both Saturday and Sunday?

6) Events C and D are such that $P(C) = 0.44$, $P(D) = 0.61$, and $P(C^C \text{ and } D) = 0.23$. Determine the value of $P(C^C \cup D)$